

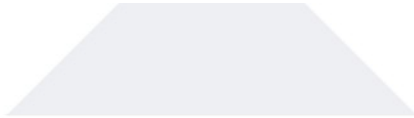
Iphone & Guardian Angels....

- First wireless computer with sensors



- Edge of the cloud technologies...



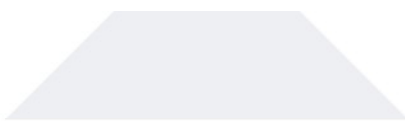


... for a smarter life: people want it...

- 2016: **500 million** edge IoT devices
- 2018: **more** edge IoT devices than Mobile phones
- 2022: **18 billion** edge IoT
- **2025: 50 Billion Edge IoT devices**
11% of the world economy
20% of energy use?!

... with billions of energy efficient IoT fog/edge devices...





... to address true global challenges
for a smarter life and planet...

- Energy, Healthcare, Environmental policies: **not sustainable**
- **More intelligence** is needed into our decision-making systems... not just more speed and capacity
- The truly disruptive topic of IoT is in the **smart cooperation between sensors and data curation quality**
- **Validity** of data for Artificial Intelligence (AI)
- More Data = **More Knowledge?** Really good for humanity without regulations on privacy and decision making empowerment?

- Iphone, Guardian Angels, Zettabyte Era & the New Ecosystem

• **Energy efficiency**

- CMOS and Beyond
 - Steep slope switches
 - 3D
- Neuromorphic computing (to overcome memory wall and *von Neumann* computing paradigm)
 - Quantum Computing
 - Energy efficient smart sensors

NOT FOR PUBLIC RELEASE



A dark brain?

- only about 3% of the neurons in the brain can be highly active at one time
- *visual processing accounts for 44% of the brain's energy consumption*

P. Lennie, Current Biology, 2003.

any disease

Nanoelectronic technologies and big challenges

**3D integration in CMOS
integrated circuits
most evolutive path for
more Moore**

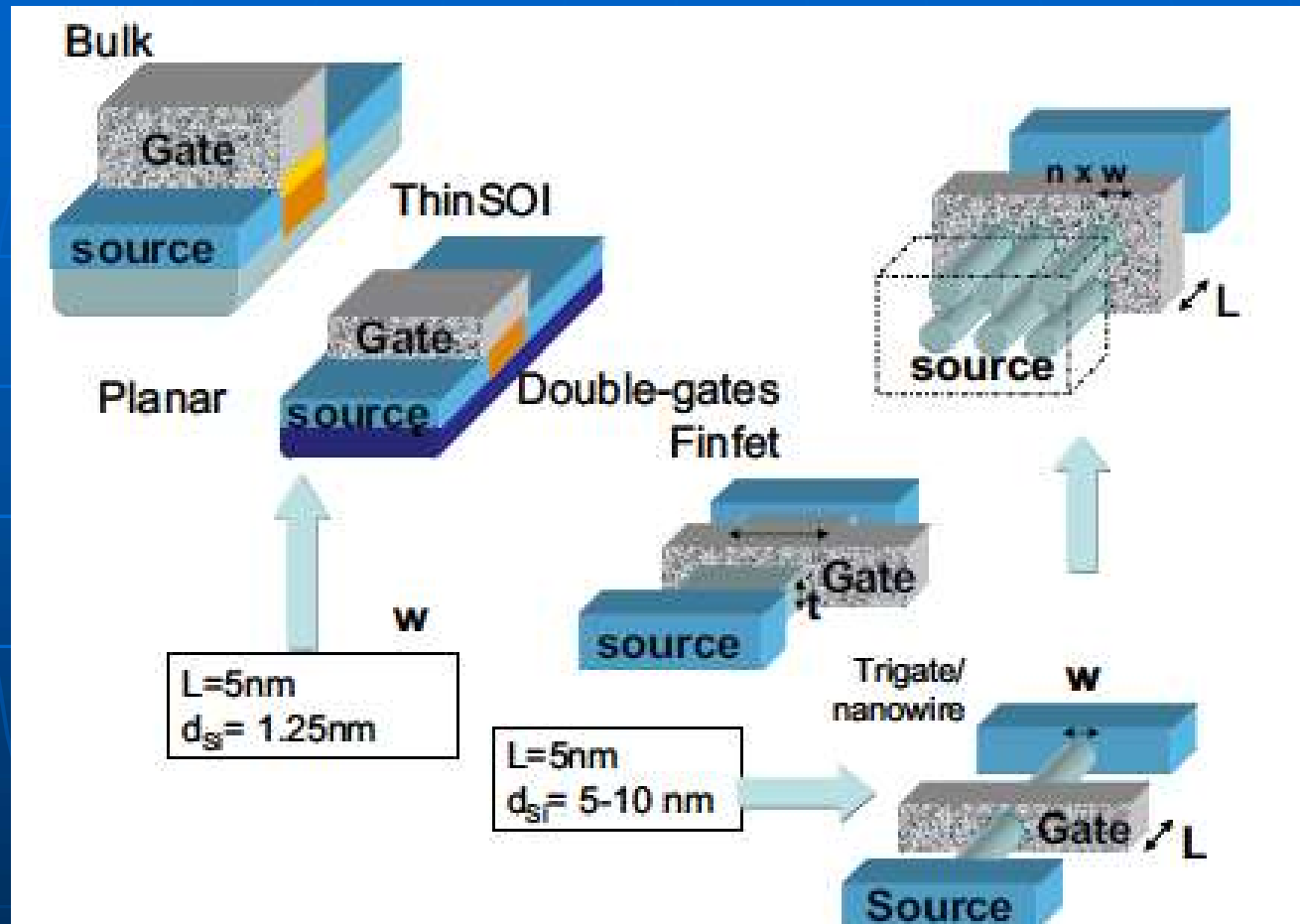
... and many possible cross contaminations

**Highly energy
efficient computing
and sensing for Big
Data, Internet of
Things and Humans**

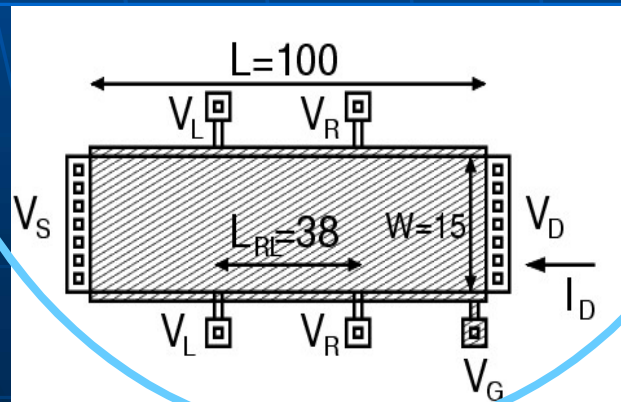
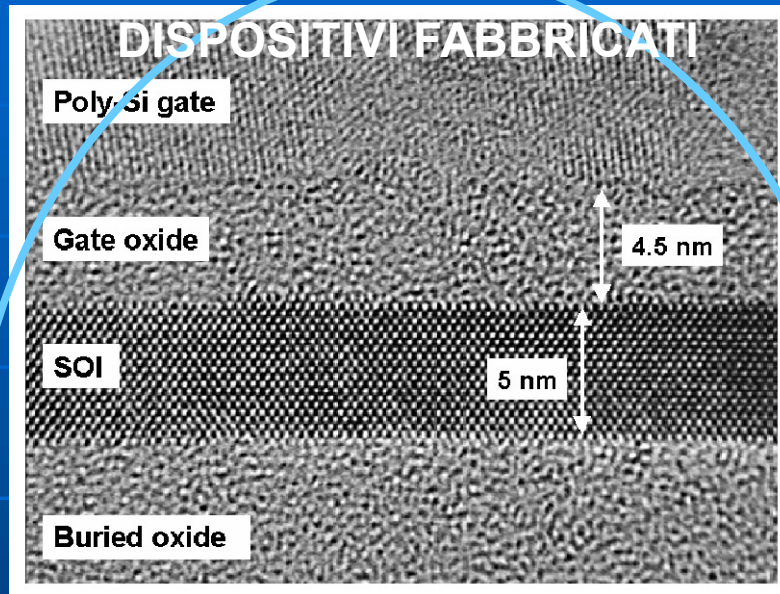
**Rebooting computing:
technology for new
computing paradigms
... neuromorphic
computing**

3D integration in CMOS integrated
circuits ... most evolutive path for more
Moore
Transistor architecture

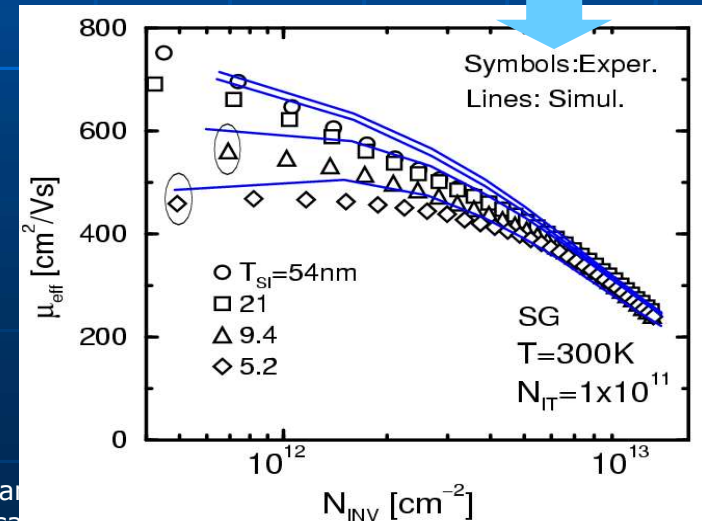
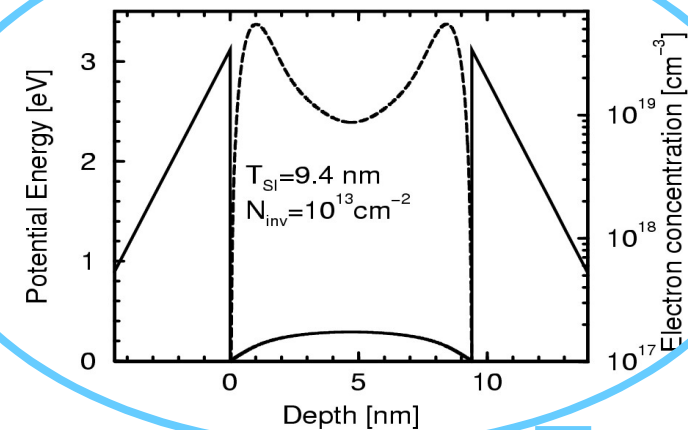
Evolution of Transistor architectures



Thin SOI MOSFETs: experimental characterization and modelling analysis at UNIUD in 2001-03



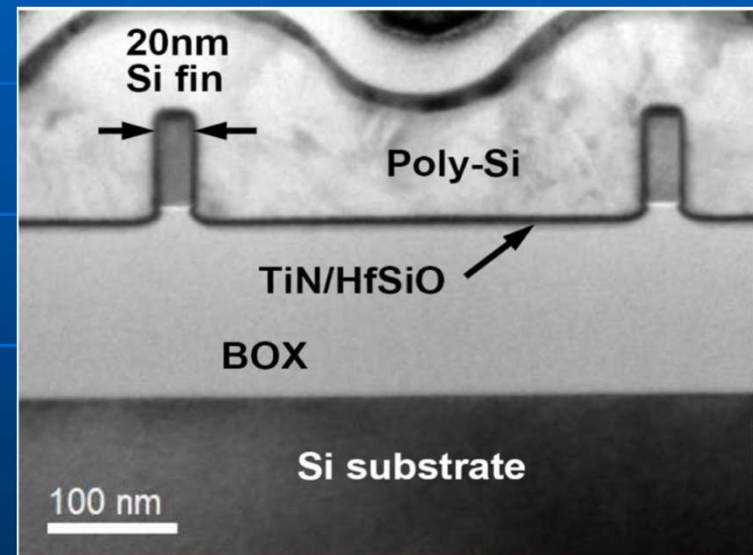
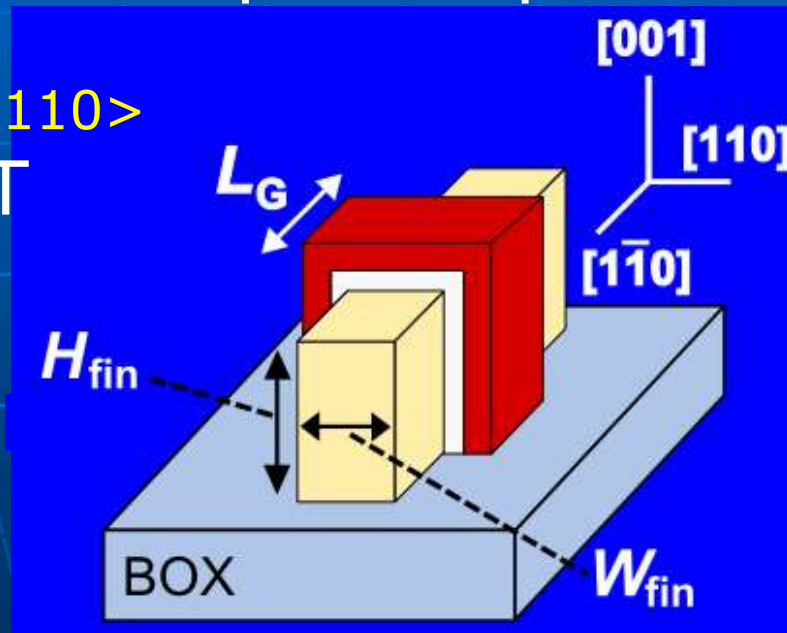
MODELLO QUANTISTICO



FinFETs: experimental characterization and modelling analysis at UNIUD in 2010-2012

- 22nm technology node
VLSI-compatible process

(110)/<110>
FinFET



Nanowire transistors: indagine modellistica a UNIUD ...

Nanowire Transistors: collaboration with Taiwan Semiconductor Manufacturing Company Europe

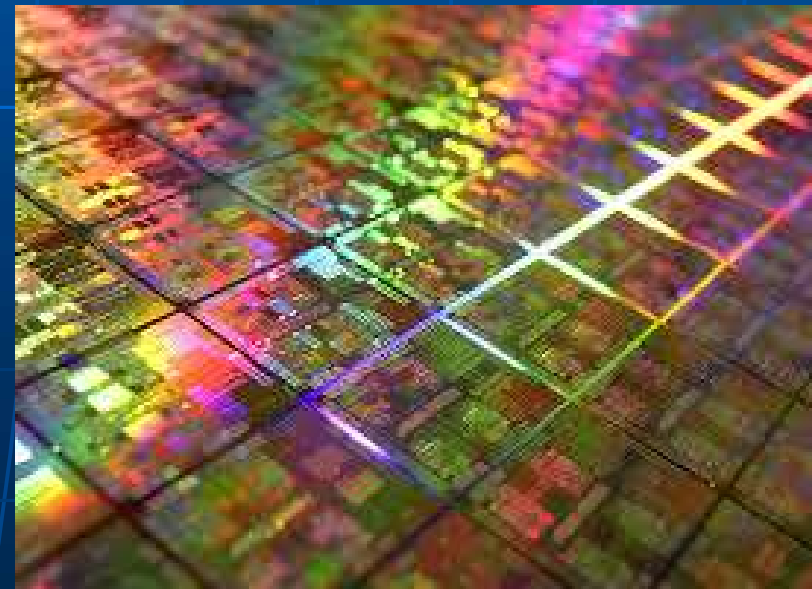
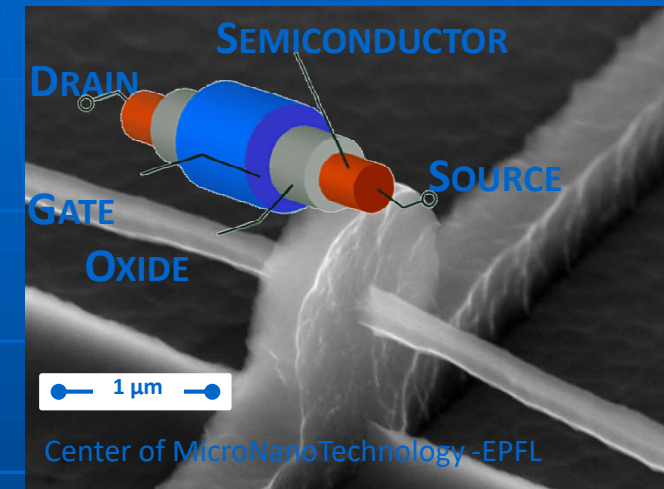
PhD student: Daniel Lizzit (ciclo XVIII)

Tutor: Prof. David Esseni

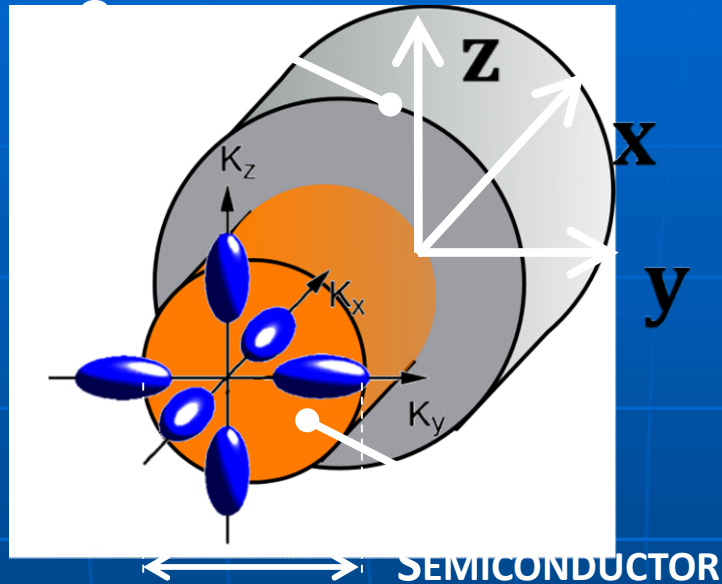
Simulation and design aspects of planar and nanowire transistors

Abstract:

This work is focused on modeling developments and design options for planar transistors and gate-all-around nanowire transistors which are among the most promising architectures for ultimate device scaling. In this framework also semiconductors alternative to silicon and belonging to the III-V group of the periodic table are investigated to improve the transistors performances.

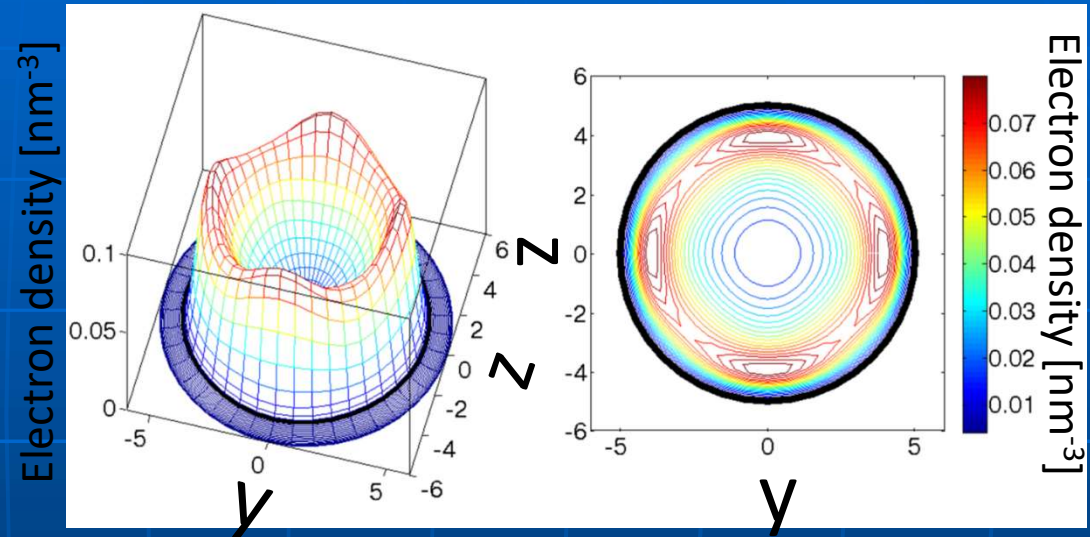


Electrostatic: Silicon GAA MOSFET



$NW_{\text{diam.}} = 10\text{nm}$

Electron density in a slice of the device

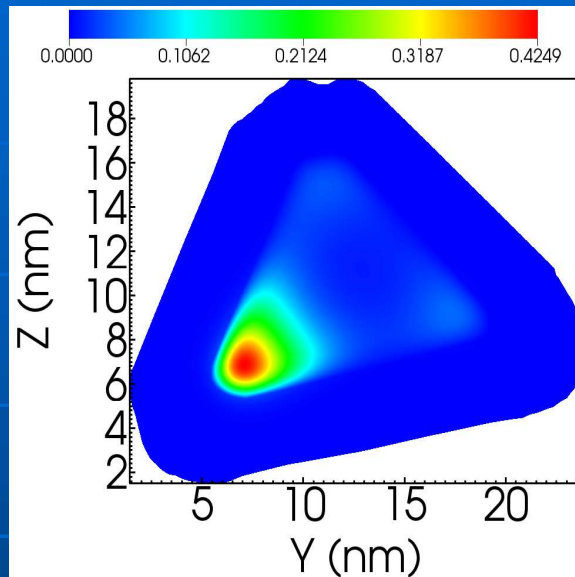


- ❑ Electron density is clearly anisotropic, reflecting the anisotropy of the electron energy dispersion relation.
- ❑ Most of TCAD simulators assumed a simplified not able to take into account valley anisotropy.

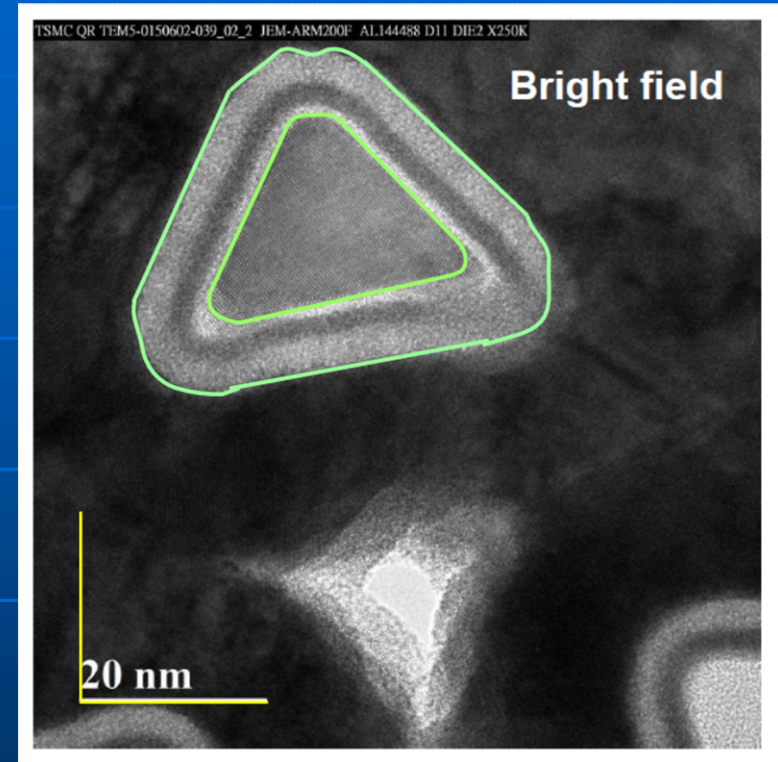
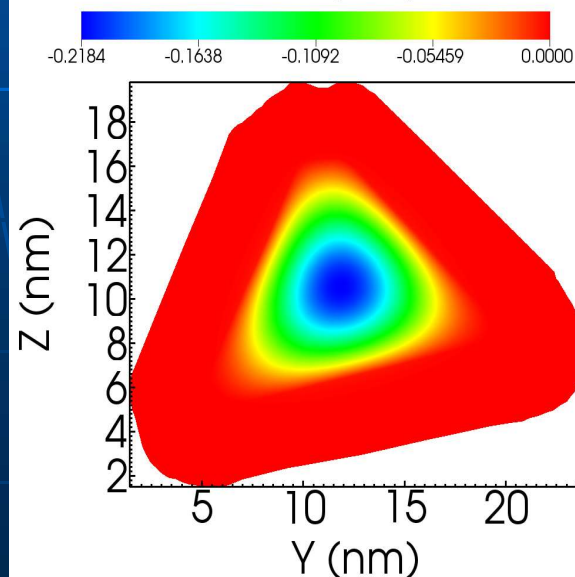
Estensione a nanowire non cilindrici

$V_g = 0.25 \text{ V}$
Charge Conc. =
 $1.79 \times 10^7 \text{ cm}^{-2}$

1st Eigen Energy



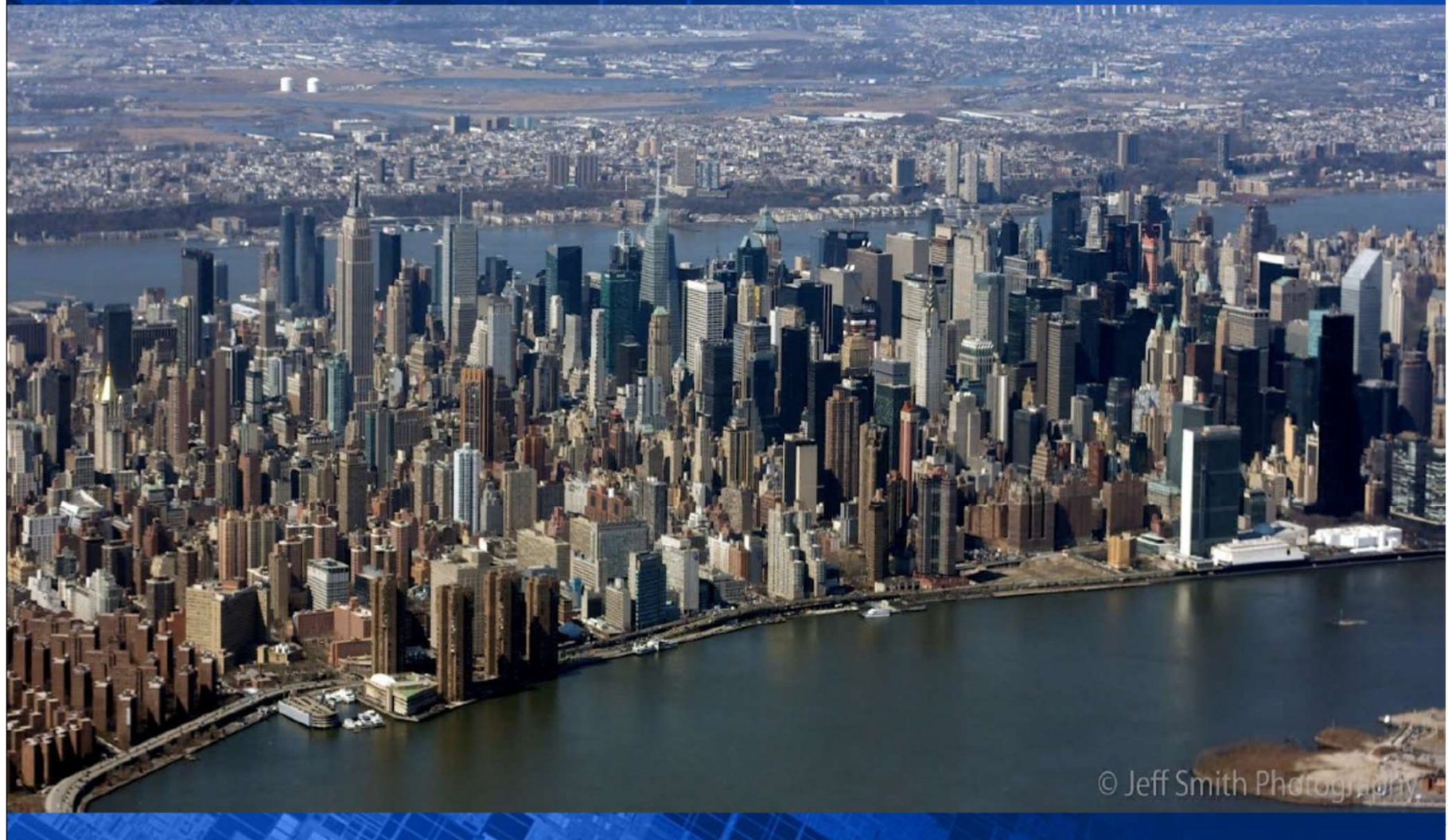
$V_g = -0.25 \text{ V}$
Charge Conc. =
 $4.89 \times 10^2 \text{ cm}^{-2}$





Towards three dimensional integrated circuit: 3D CMOS technologies

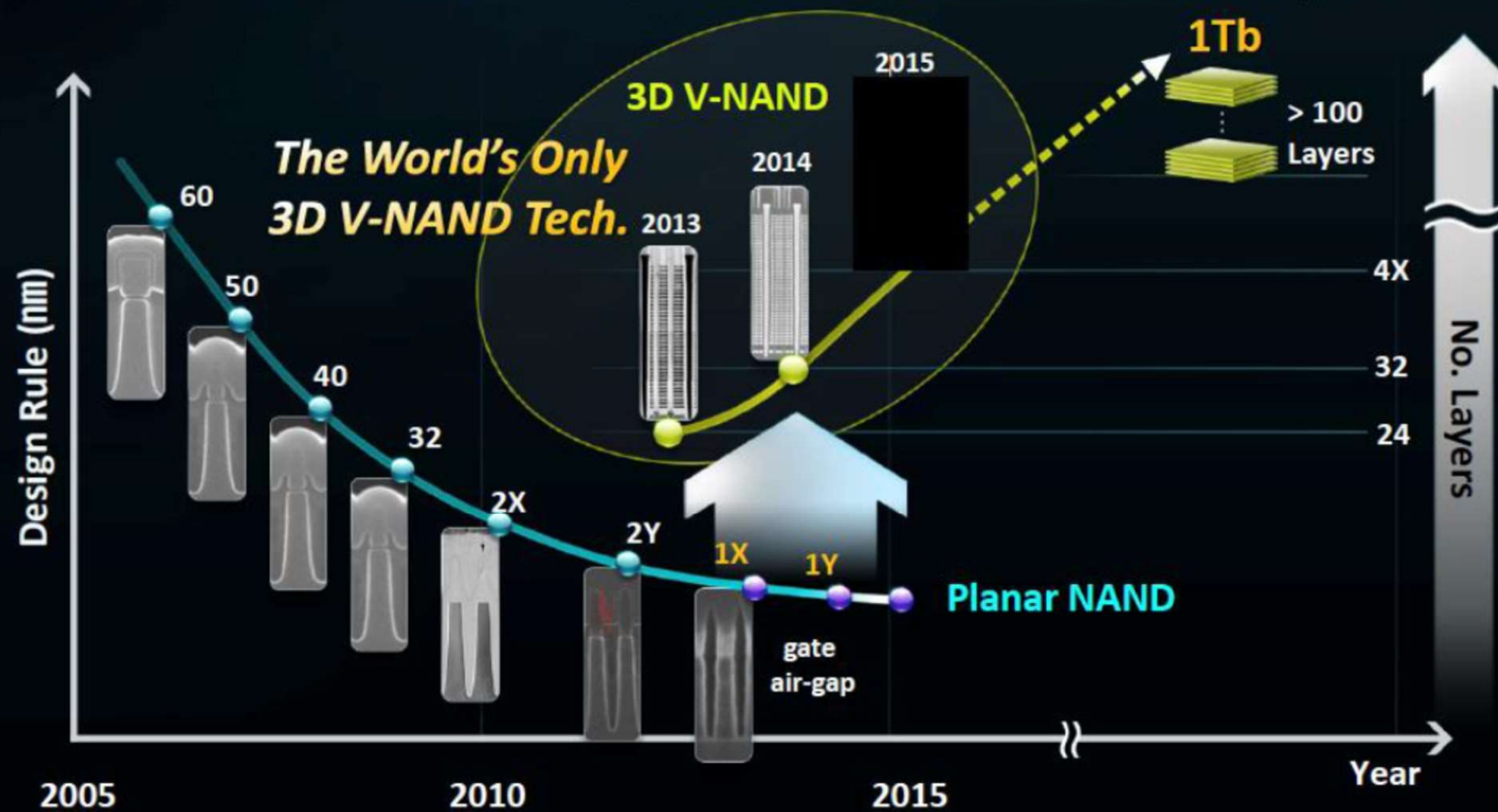
3D Manhattan Solution



Circuiti e sistemi: seminario nano-
transistori e nano-elettronica

NAND Flash Technology Outlook

Planar NAND will reach a limit, while V-NAND will continue beyond 1Tb



Kinam Kim, IEEE Fellow, President Samsung Electronics

Monolithic Heterogeneous Computing Architecture



3D Resistive RAM
Massive storage

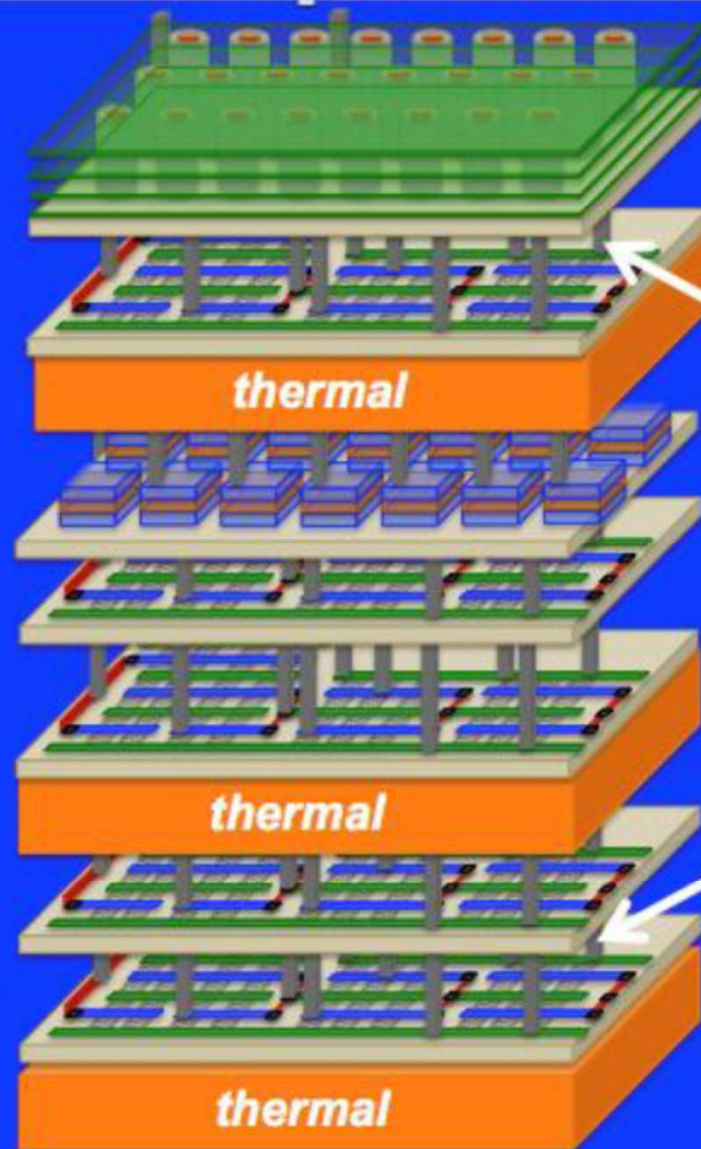


1D CNFET, 2D FET
Compute, RAM access

STT MRAM
Quick access

1D CNFET, 2D FET
Compute, RAM access

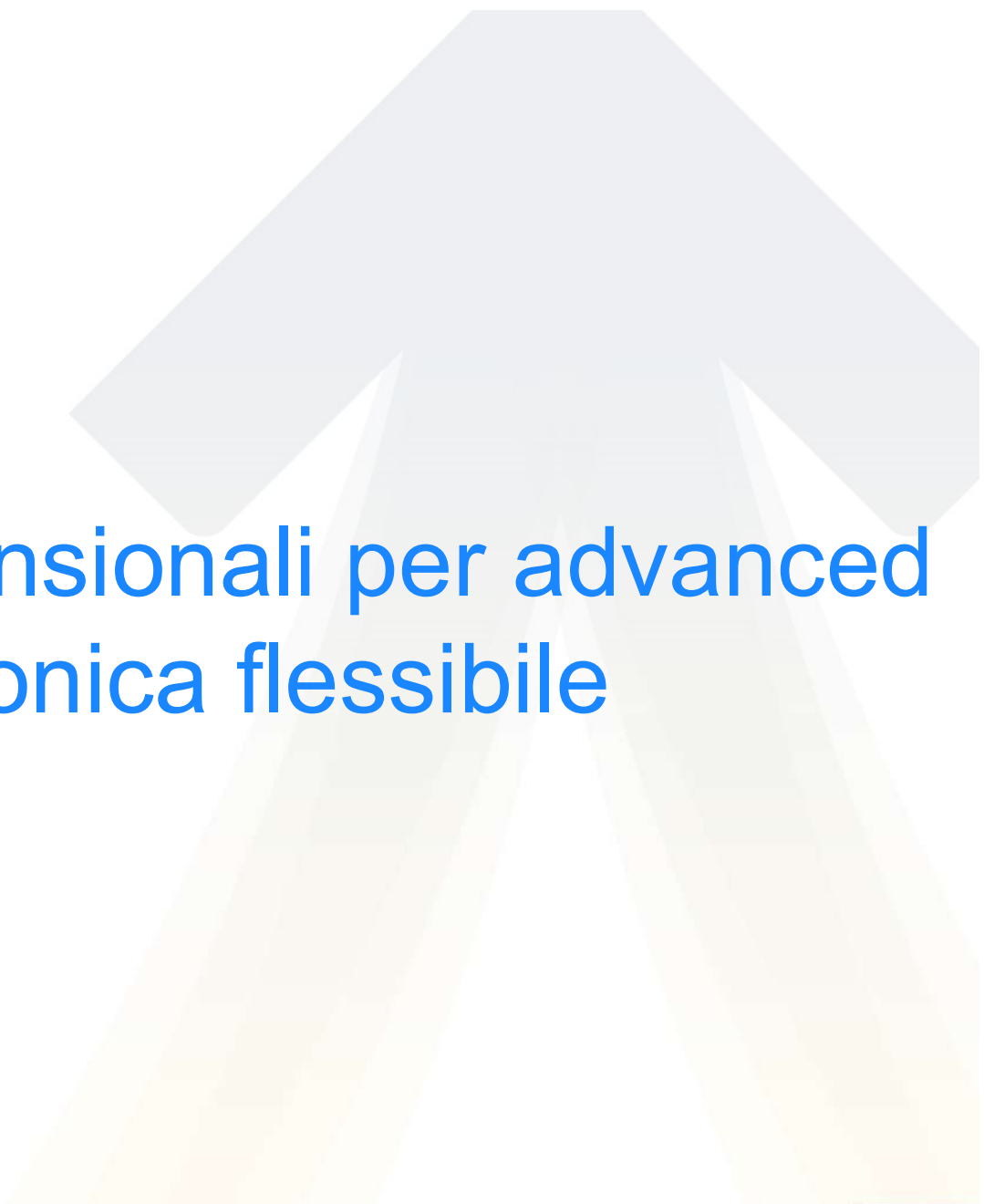
1D CNFET, 2D FET
Compute, Power, Clock



No TSV

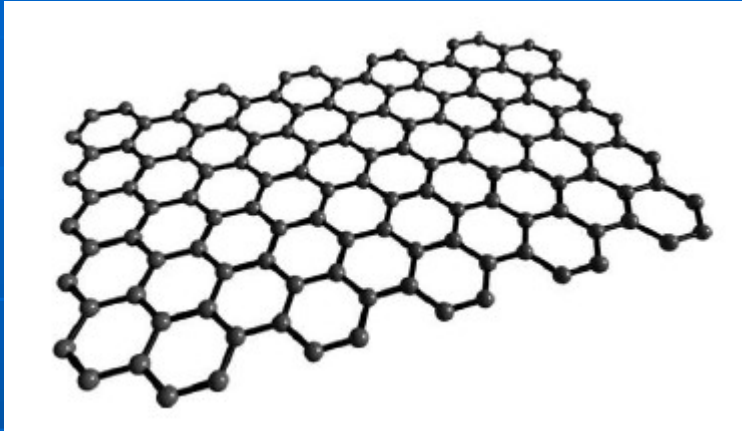
Dense
nano-scale
inter-layer vias

Silicon
compatible

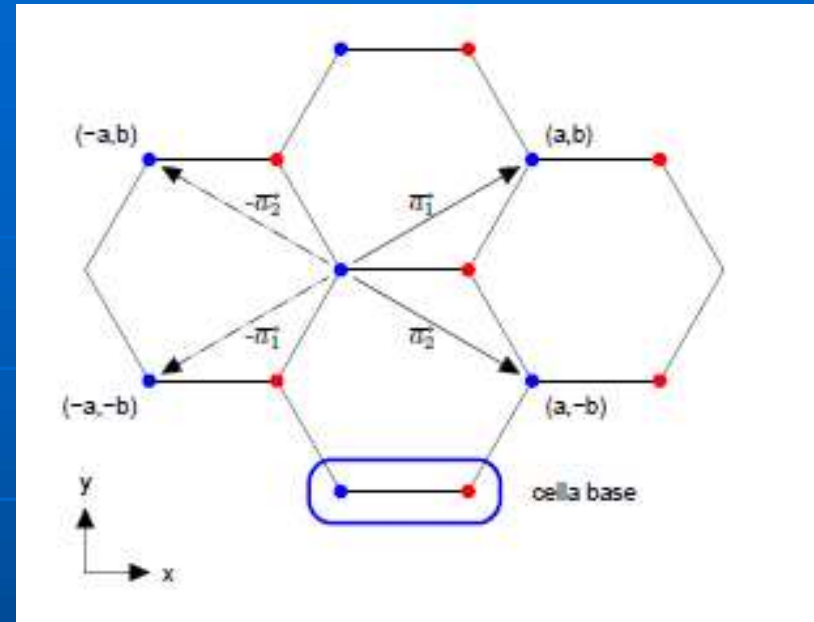
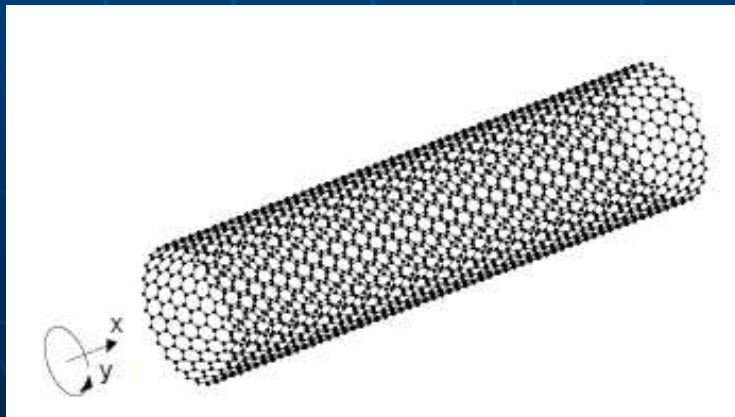


Materiali bidimensionali per advanced CMOS ed elettronica flessibile

Il materiale grafene

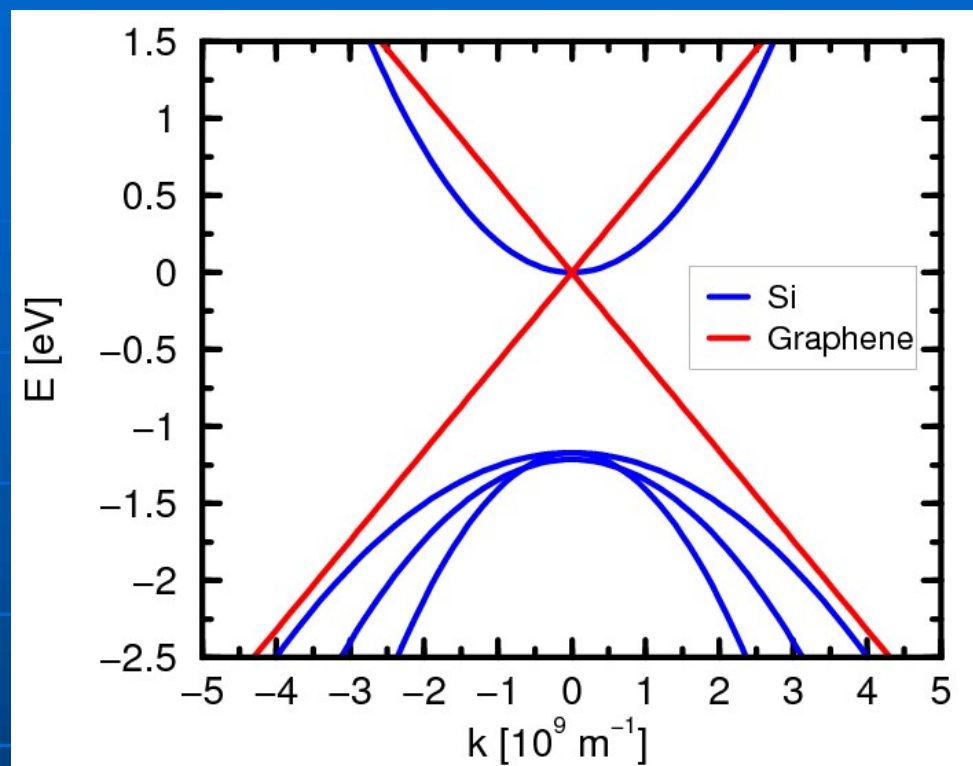


- Foglio di grafene come materiale naturalmente bidimensionale



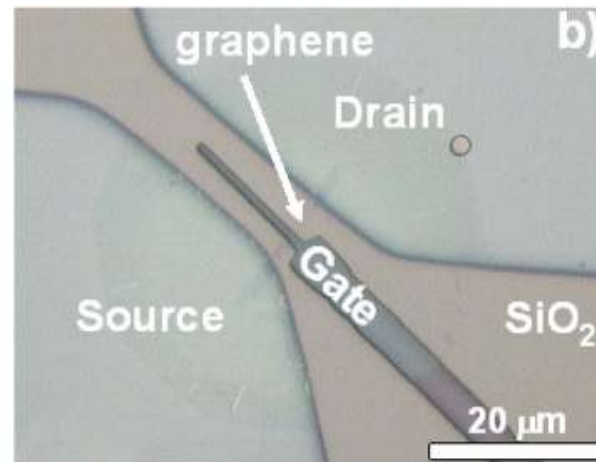
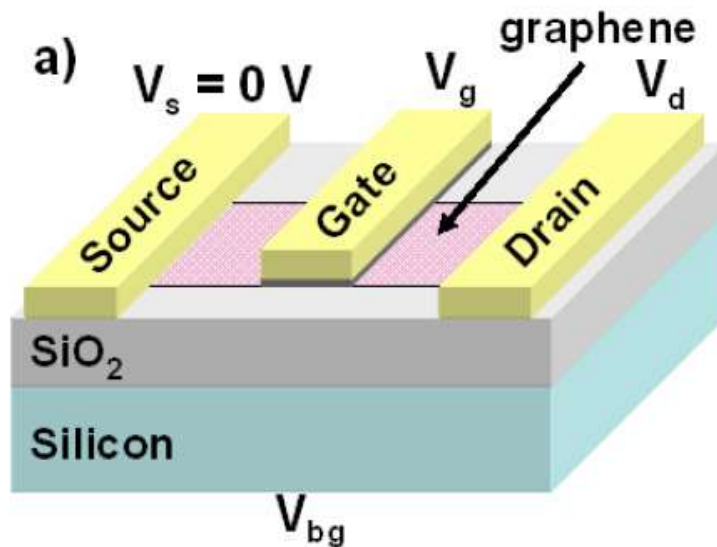
- Reticolo cristallino di un foglio di grafene
- Cella elementare biatomica
- Nanotubo ottenuto arrotolando il grafene secondo opportune chiralità

Grafene è semimetallo: energy GAP è nullo



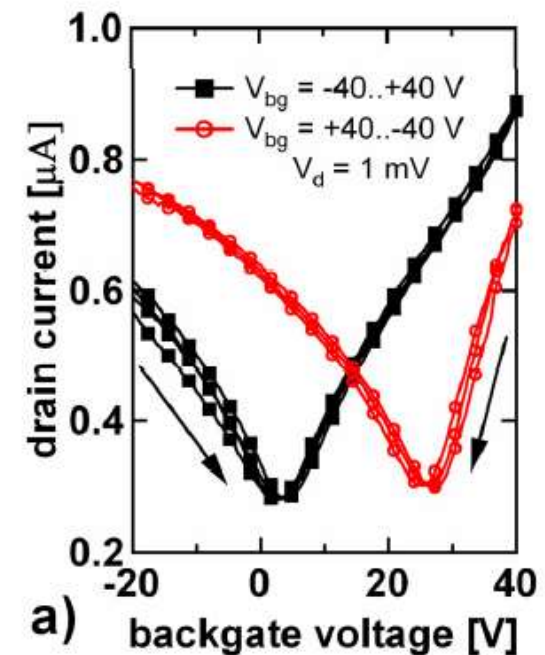
- Relazione E-k lineare invece che parabolica
→ Massa efficace virtualmente nulla
Mobilità elevata fino a $10^6 \text{ cm}^2/(\text{Vs})$ [Geim 2010]

Transistore al grafene

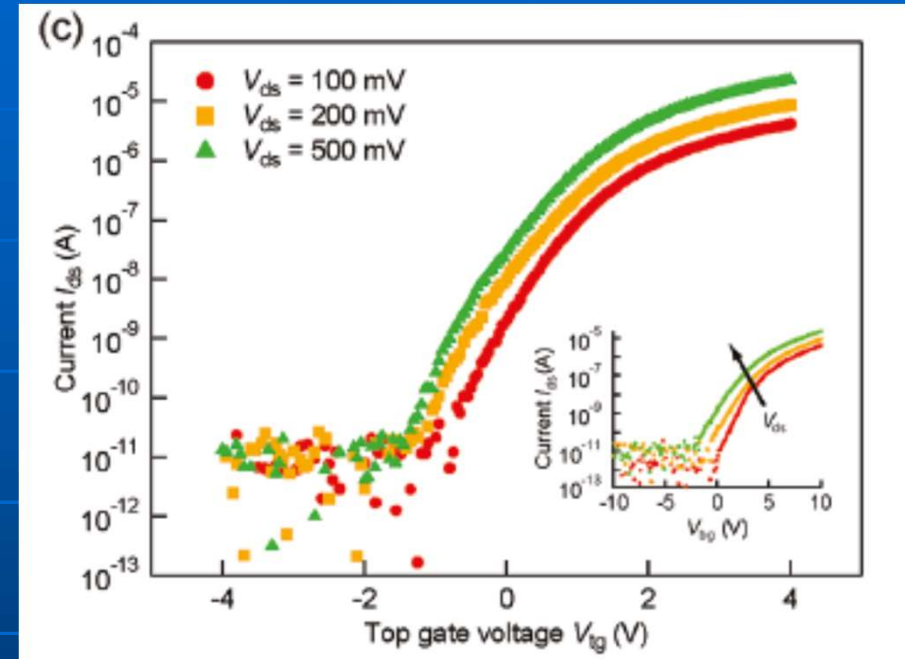
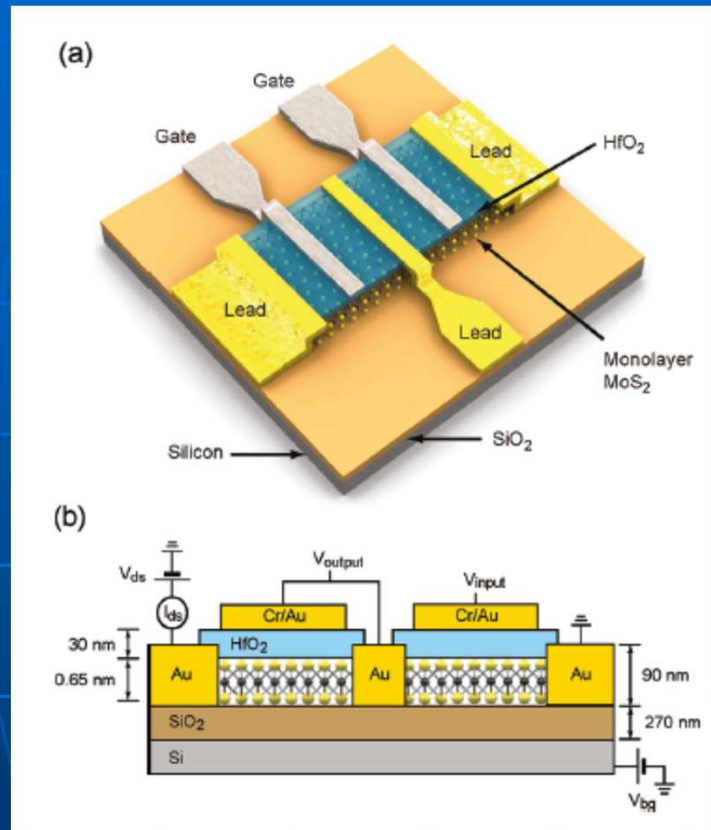


- Assenza di un Energy-Gap
→ Rapporto I_{ON}/I_{OFF} basso
- 1) Comportamento ambipolare nelle caratteristiche $I_{ds}-V_{gs}$
- 2) Pessima saturazione nelle caratteristiche di uscita $I_{ds}-V_{ds}$

Circuiti e sistemi: seminario nano-transistori e nano-elettronica



Transistori con 2D gapped semiconductors (after B.Radisavljevic et al. ACS Nano 2011)



- MoS₂ ha un energy gap di circa $E_{GAP}=1.8\text{eV}$
- Grazie al Energy-Gap i transistori hanno I_{ON}/I_{OFF} di molti ordini di grandezza !!

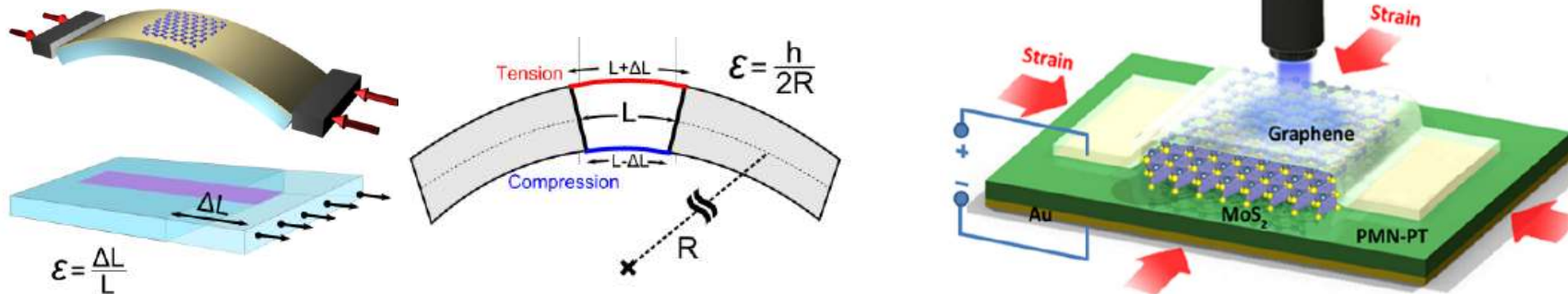
2D gapped semiconductors per sensori ultra-sottili (after M.Hosseini D.Esseni, ESSDERC, Sept. 2015)



University of
Tehran



- **Very high stretchability: sustain strain > 10%** [D.-M. Tang *et al.* Nature Comm. 2014; A.Castellanos Gomez *et al.* Nano Res. 2015]
- Several techniques to apply either uniaxial or biaxial strain



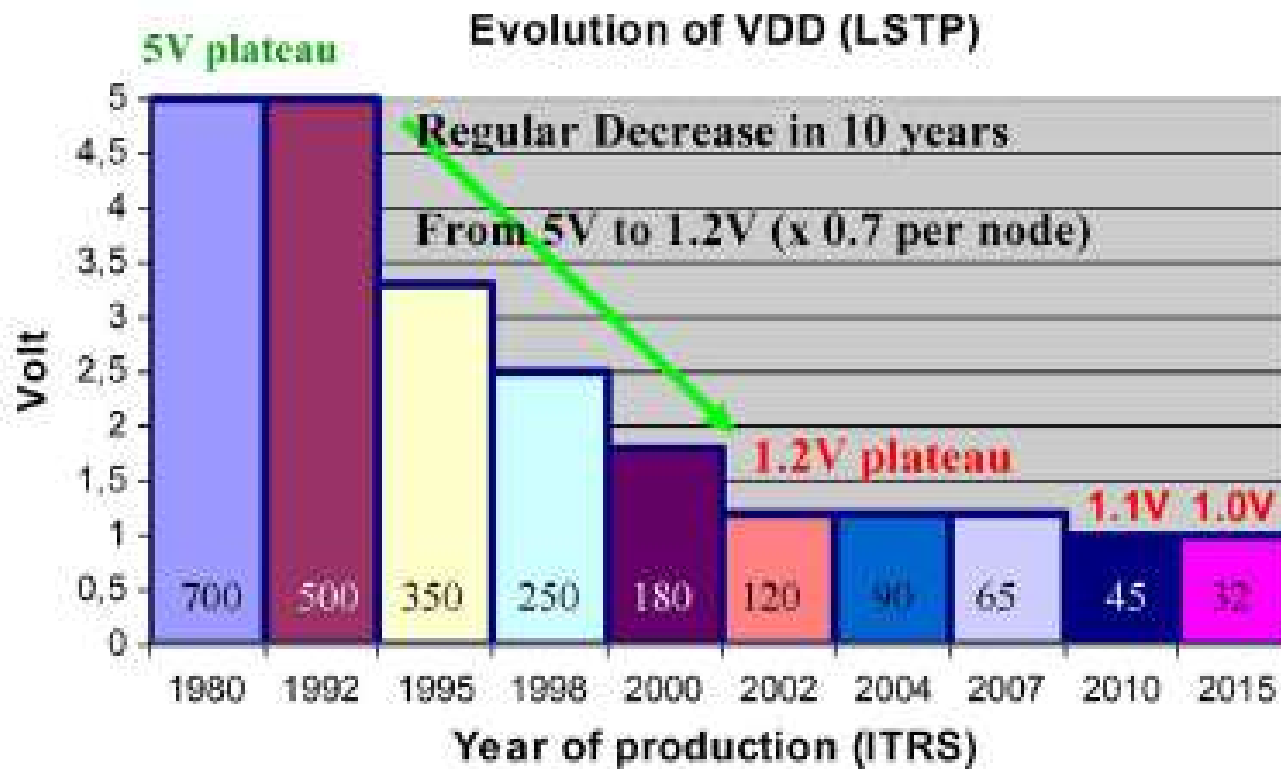
Straining technique	Type of strain		Max. strain	Reference
Bending or elongating of a flexible substrate	Uni.	Homogeneous	2.4 % 0.5 % 0.8 %	Conley, et al., Nano Lett. 13, 3626 (2013). He, et al., Nano Lett. 13, 2931 (2013). Zhu, et al., Phys. Rev. B 88, 121301 (2013).
Piezoelectric stretching	Bi.	Homogeneous	0.2 %	Hui, et al., ACS Nano 7, 7126 (2013).
Exploiting the thermal expansion mismatch	Bi.	Homogeneous	0.23 %	Plechinger, et al. 2D Mater. 2, 015006 (2015).
Controlled wrinkling	Uni.	Inhomogeneous	2.5 %	C-Gomez,. et al., Nano Lett. 13, 5361 (2013)

Nanoelectronic technologies and big challenges

**Highly energy
efficient computing
and sensing for Big
Data, Internet of
Things and Humans**

Power crisis and voltage scaling

$$\text{Power} = P_{\text{sw}} (f_{\text{clk}} C_{\text{eff}} V_{\text{DD}}^2) + (1 - P_{\text{sw}})(I_{\text{OFF}} V_{\text{DD}})$$



V_{DD} has stagnated at around 1V

Is Dark Silicon Useful?

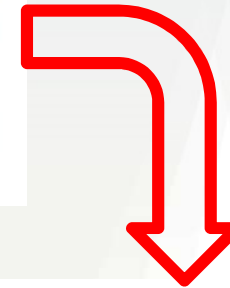
Harnessing the Four Horsemen of the Coming Dark Silicon Apocalypse

Michael B. Taylor

DAC 2012, June 3-7, 2012, San Francisco, California, USA.

ABSTRACT

Due to the breakdown of Dennardian scaling, the percentage of a silicon chip that can switch at full frequency is dropping exponentially with each process generation. This utilization wall forces designers to ensure that, at any point in time, large fractions of their chips are effectively dark or dim silicon, i.e., either idle or significantly underclocked.



6. THE DEUS EX MACHINA HORSEMAN

comes out of nowhere to save the day. In the case of dark silicon, one Deus Ex Machina would be a breakthrough in semiconductor devices. However as we shall see, the breakthroughs that would be required would have to be quite fundamental – in fact most likely would require us to build transistors out of devices other than MOSFETs. The rea-

Nanoelectronic Devices and Circuits with High Energy Efficiency for Internet-of-Things Applications

David Esseni

DPIA, University of Udine, Via delle Scienze 206, 33100 Udine, Italy; email: david.esseni@uniud.it

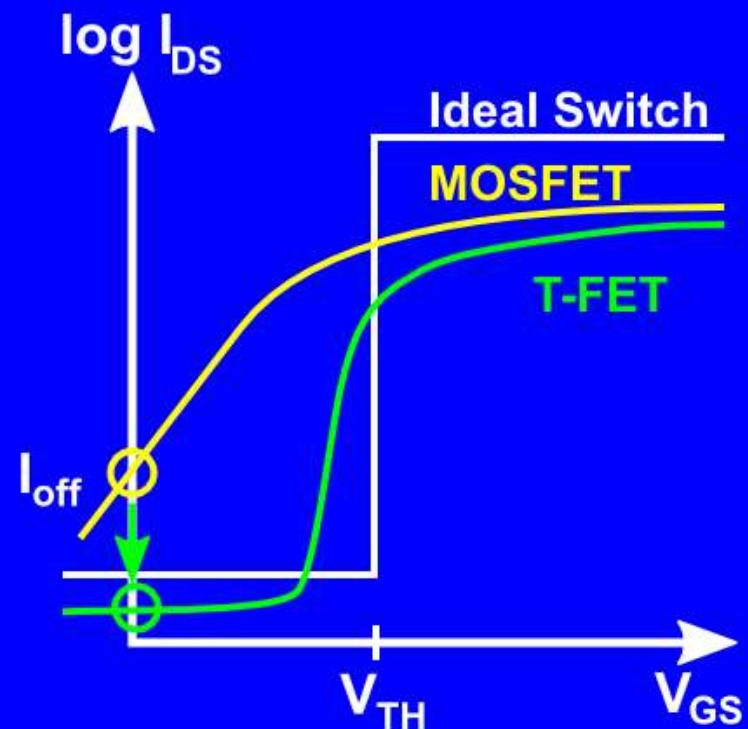
13 dicembre 2021

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Small swing FETs: Tunnel FETs

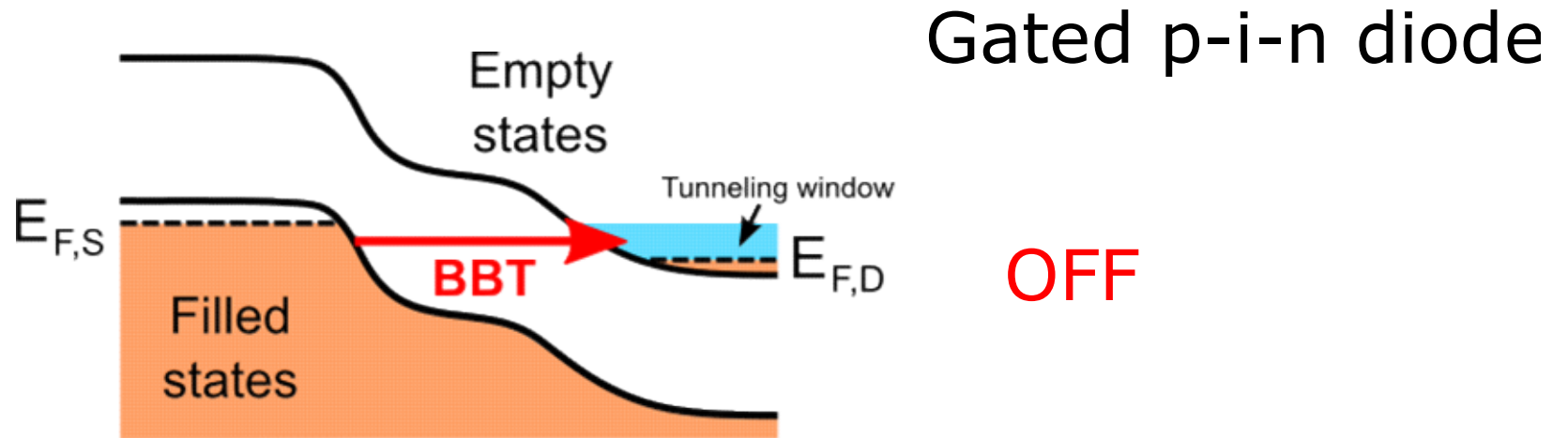
- Inverse sub-threshold slope (SS) below 60mV/dec at room T

➔ Potential improvement of $[I_{ON} / I_{OFF}]$ tradeoff compared to MOSFETs



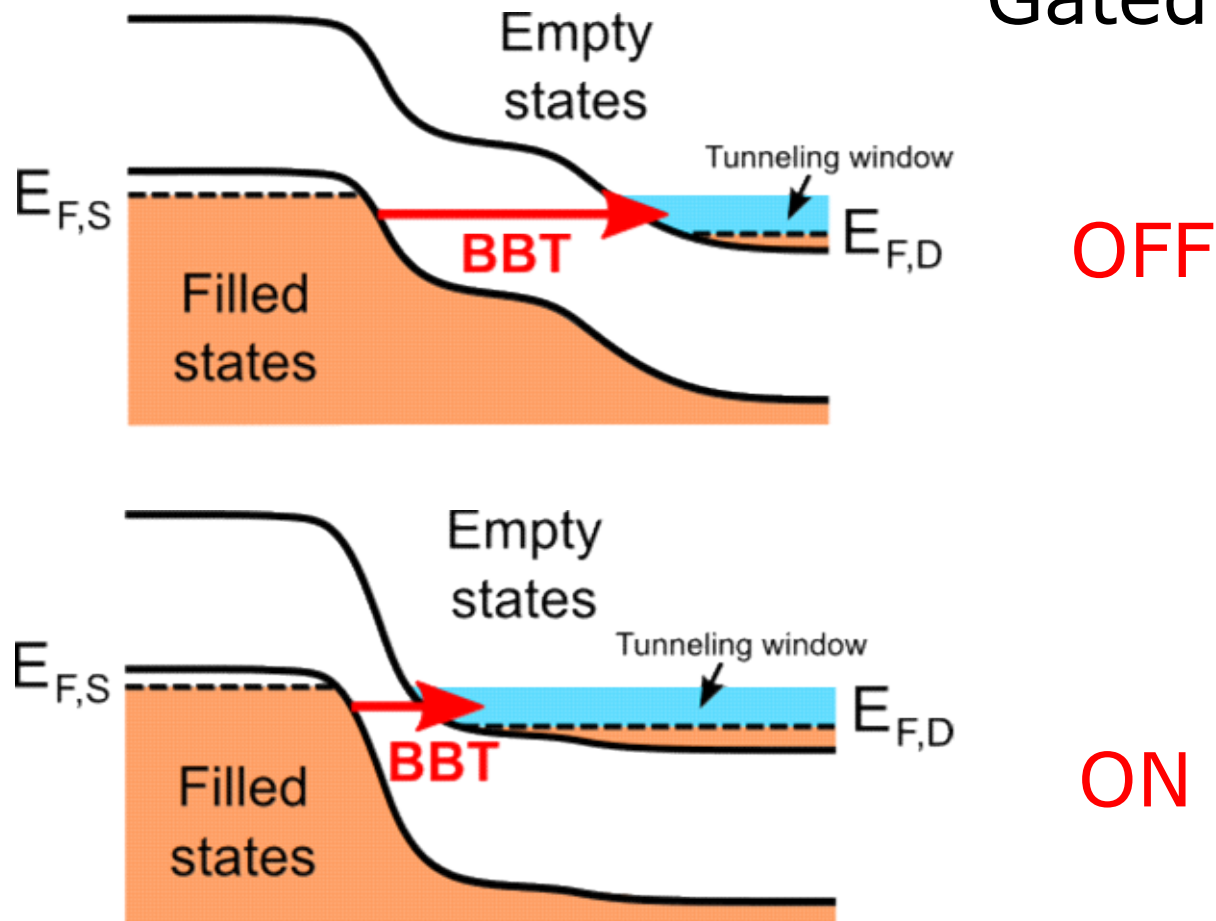
- Possible device concepts:
 - Tunnel FETs based on band-to-band-tunnelling
 - Negative capacitance transistors, NC-FETs

Tunnel-FETs: working principle



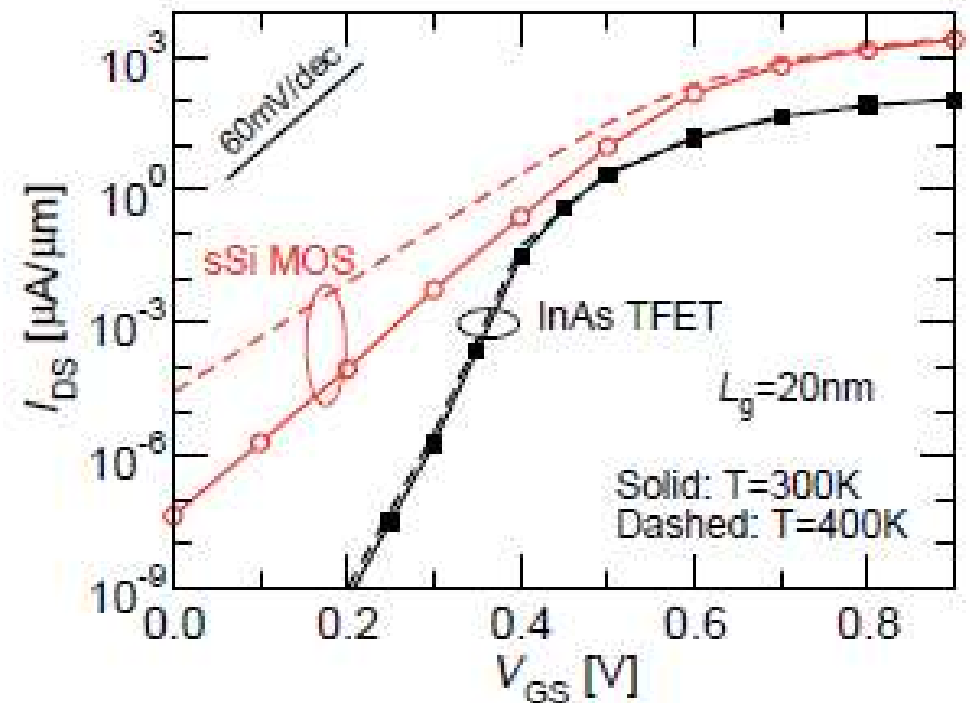
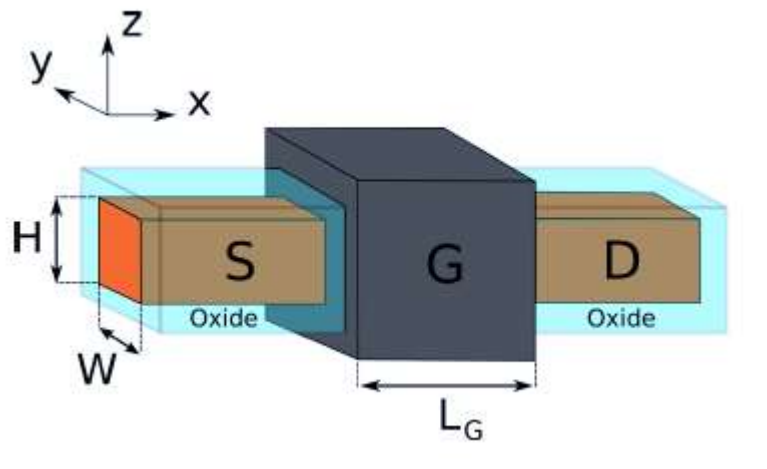
Tunnel-FETs: working principle

Gated p-i-n diode



InAs nanowire Tunnel FETs

(F. Conzatti et al. IEDM 2011)

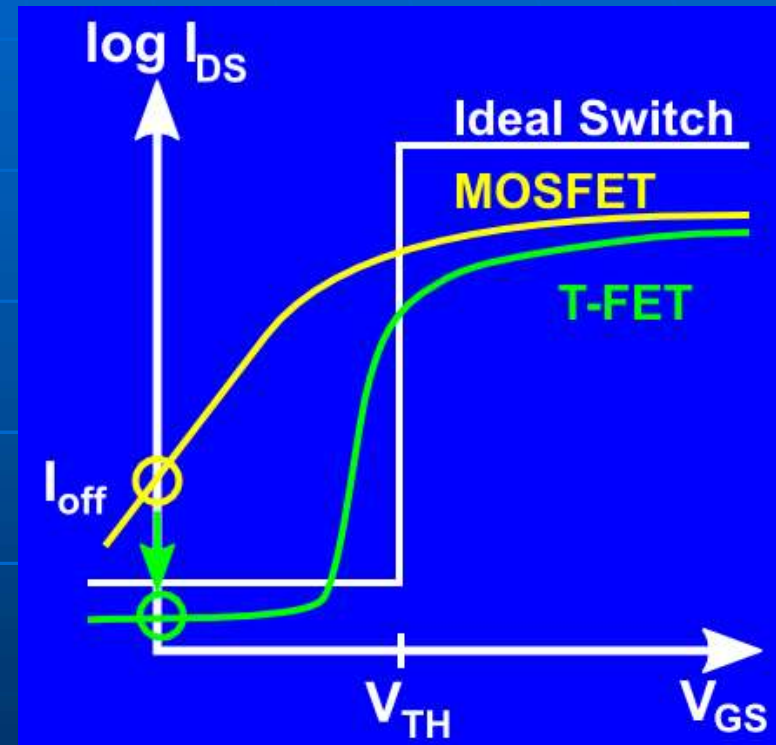


- Simulazioni con trasporto balistico di InAs Tunnel FETs confrontati con MOSFET convenzionali
- Migliore characteristics di sottosoglia

Small swing FETs: Negative Capacitance FETs

- Inverse sub-threshold slope (SS) below 60mV/dec at room T

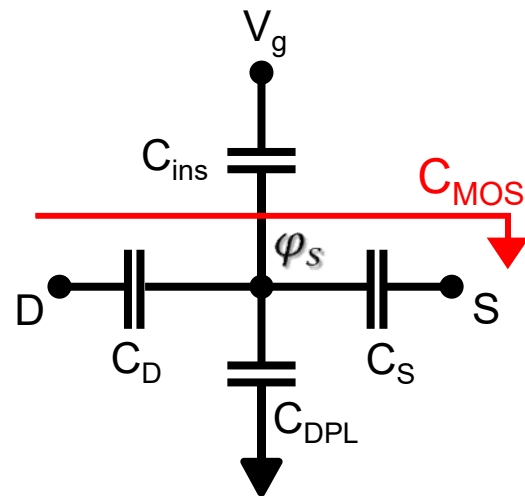
➔ Potential improvement of $[I_{ON} / I_{OFF}]$ tradeoff compared to MOSFETs



- Possible device concepts:
 - Tunnel FETs based on band-to-band-tunnelling
 - **Negative capacitance transistors, NC-FETs**

NC-FET concept [s. Salahuddin et al., Nano Letters, 2008]

The **Negative Capacitance** concept can in principle reduce S below 60mV/dec



$$S = \frac{\partial V_g}{\partial \text{Log}(I_{DS})} = \left[\frac{1}{m} \right] \left(\frac{\partial \phi_s}{\partial \text{Log}(I_{DS})} \right)$$

$$\Rightarrow m = \left[\frac{\partial \phi_s}{\partial V_G} \right] = \frac{C_{ins}}{C_{ins} + C_{MOS}} < \mathbf{1 \text{ in conventional MOS with } C_{ins} > 0}$$



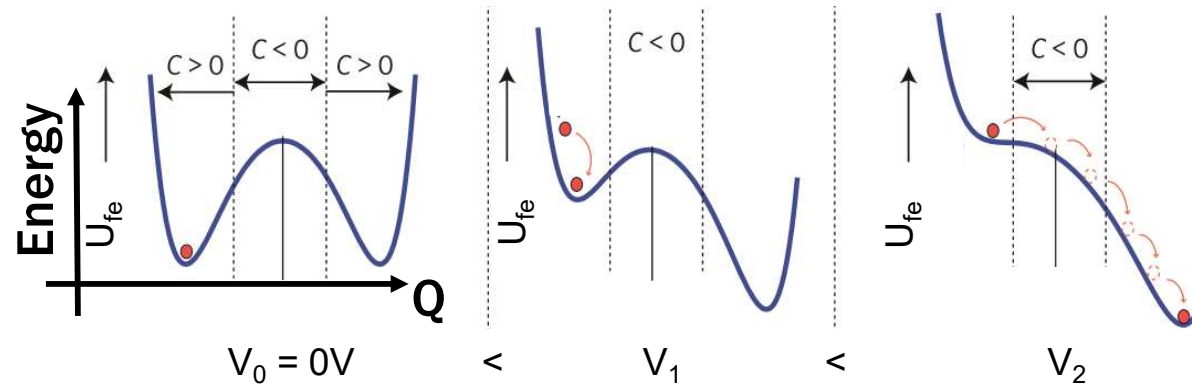
Ferroelectric materials (BaTiO₃, Si:HfO₂, Hafnium Zirconium Oxide - HZO) show a negative capacitance **C_{FE}** branch, such that charge increases for decreasing voltage:

$$C_{ins} = C_{FE} = - |C_{FE}|$$

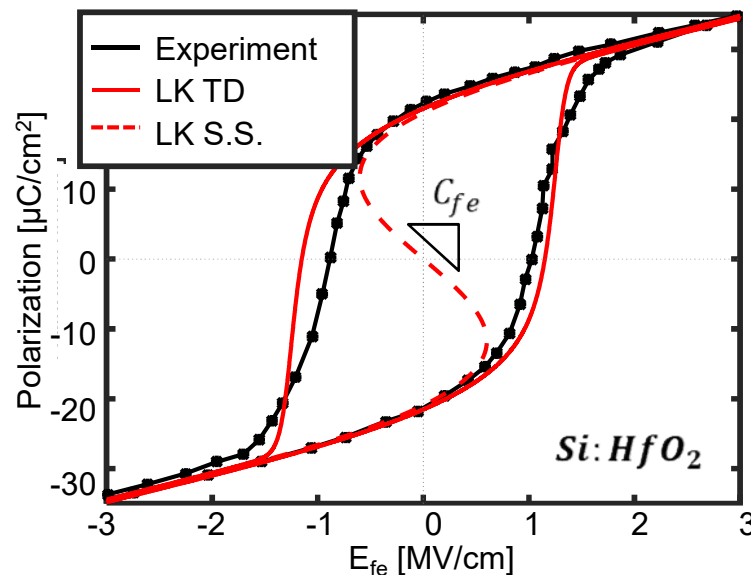
$$m = \frac{|C_{FE}|}{|C_{FE}| - C_{MOS}} > 1 \quad \mathbf{BUT \text{ POSITIVE !!}} \quad \Rightarrow \quad \mathbf{S < 60 \text{ mV/dec}}$$

NC-FET: where does the negative C_{FE} stem from ?

Ferroelectrics are usually described by the Landau-Khalatnikov Equation (LKE):

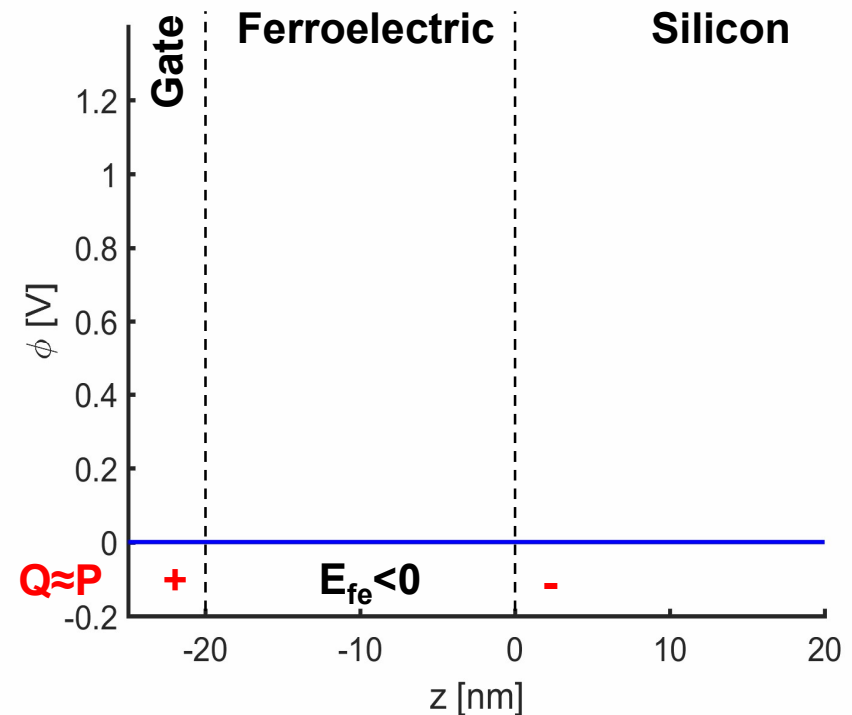
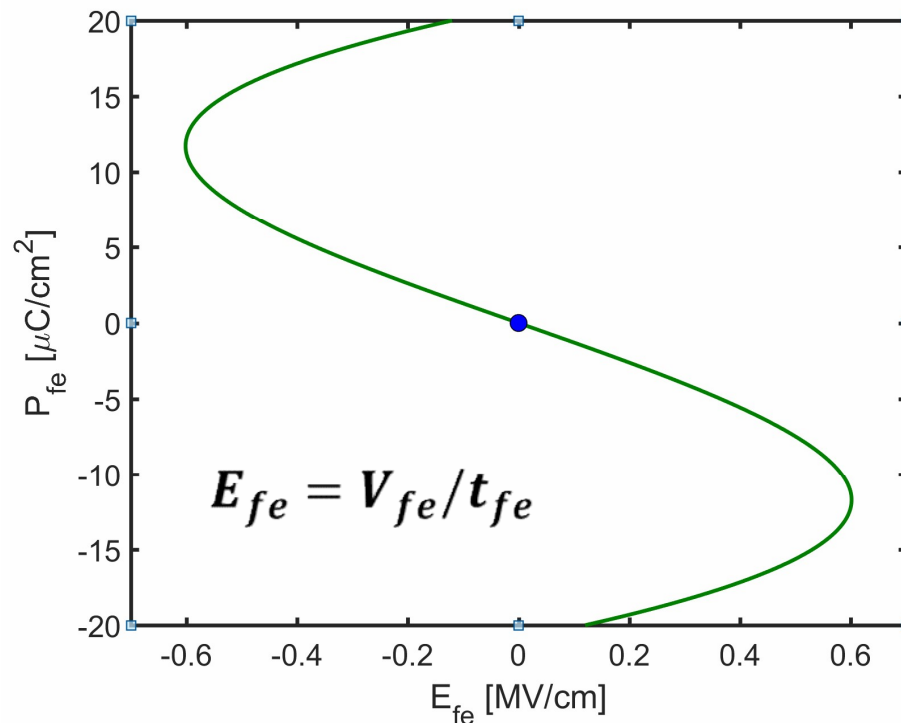
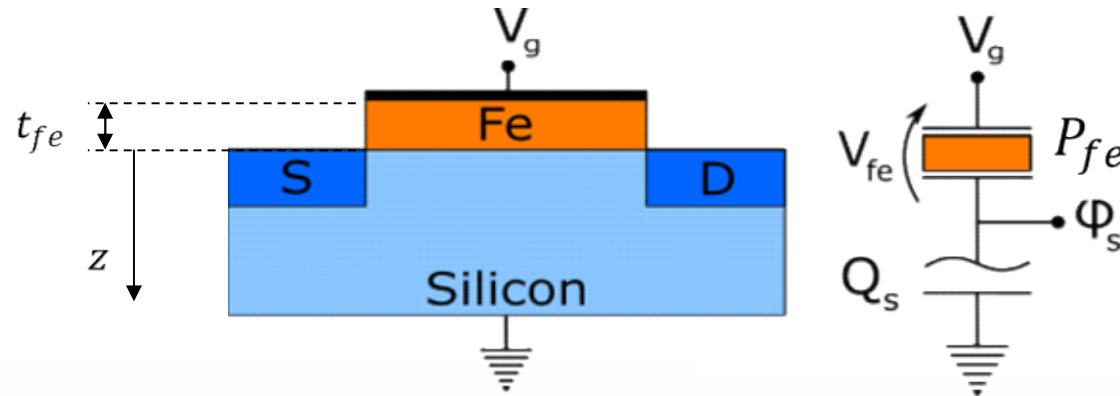


LKE is an Energy-Minimization Approach



- 1) Ferroelectric alone is unstable in small E_{FE} region and it has hysteric behavior
- 2) Ferroelectric with positive capacitance in series should can be stabilized in small E_{FE} region (around $E_{FE}=0$) and exhibit a negative capacitance behavior $C_{FE} < 0$ [S. Salahuddin *et al.*, Nano Letters, 2008]

Negative capacitance FETs: voltage gain $\left| \frac{\partial \phi_s}{\partial V_g} \right| > 1$



Negative capacitance FETs

Is there experimental evidence of sub-threshold swing below 60mV/dec in NC-FETs ?

[P.Sharma *et al.*, VLSI Symp. 2017]

After P.Sharma et al., VLSI Symp. 2017

Gate last process with $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ (HZO) as ferroelectric (FE) dielectric in a metal/ferroelectric/insulator/semiconductor (MFIS) configuration

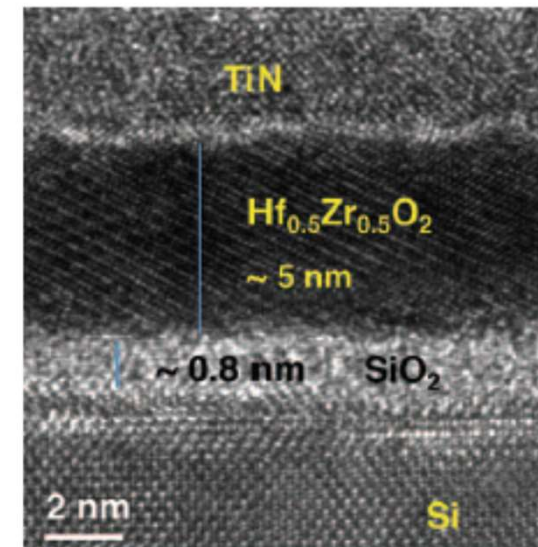
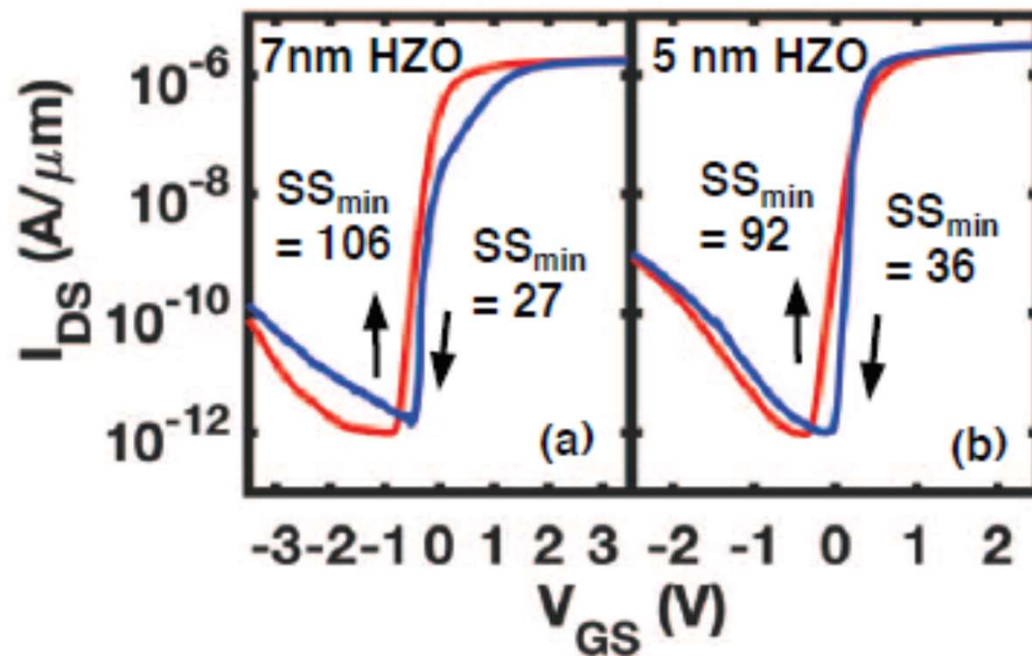


Fig. 2: HRTEM cross-section of MFIS gate stack with 5 nm HZO.

- Long channel FETs (around $2\mu\text{m}$)
- Asymmetric and hysteretic I_{DS} - V_{GS} characteristics
- Measurements at low frequency and still too large voltages

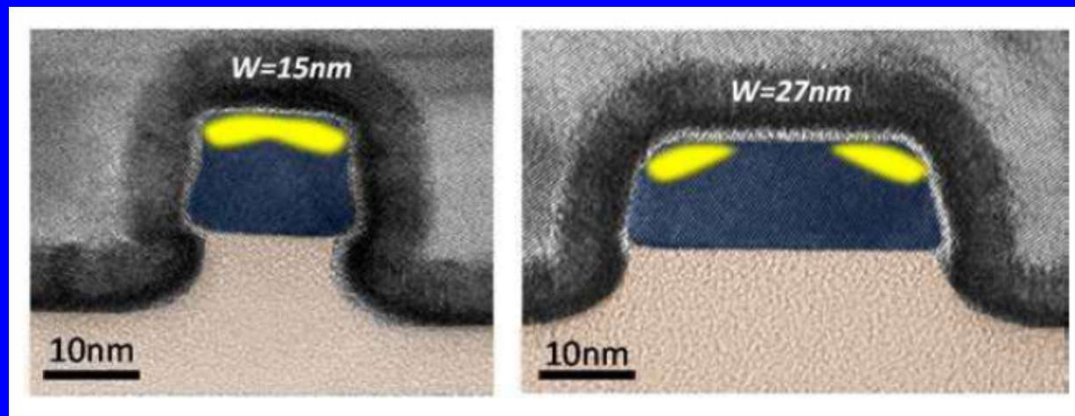
Nanoelectronic technologies and big challenges

**SOI CMOS technology for
quantum information
processing**
[S. de Franceschi *et al.*
IEDM 2016]

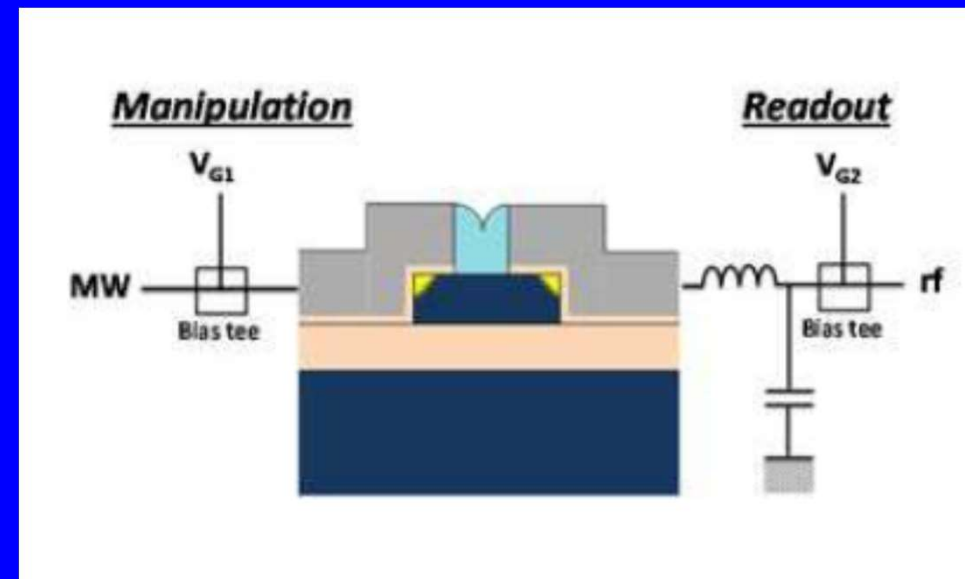
**Rebooting computing:
technology for new
computing paradigms**
... **quantum computing,**
neromorph. computing

SOI technology for quantum information processing

[S. de Franceschi *et al.* IEDM 2016]



Efforts are going on at LETi and IMEC to exploit advanced CMOS technologies for quantum computing

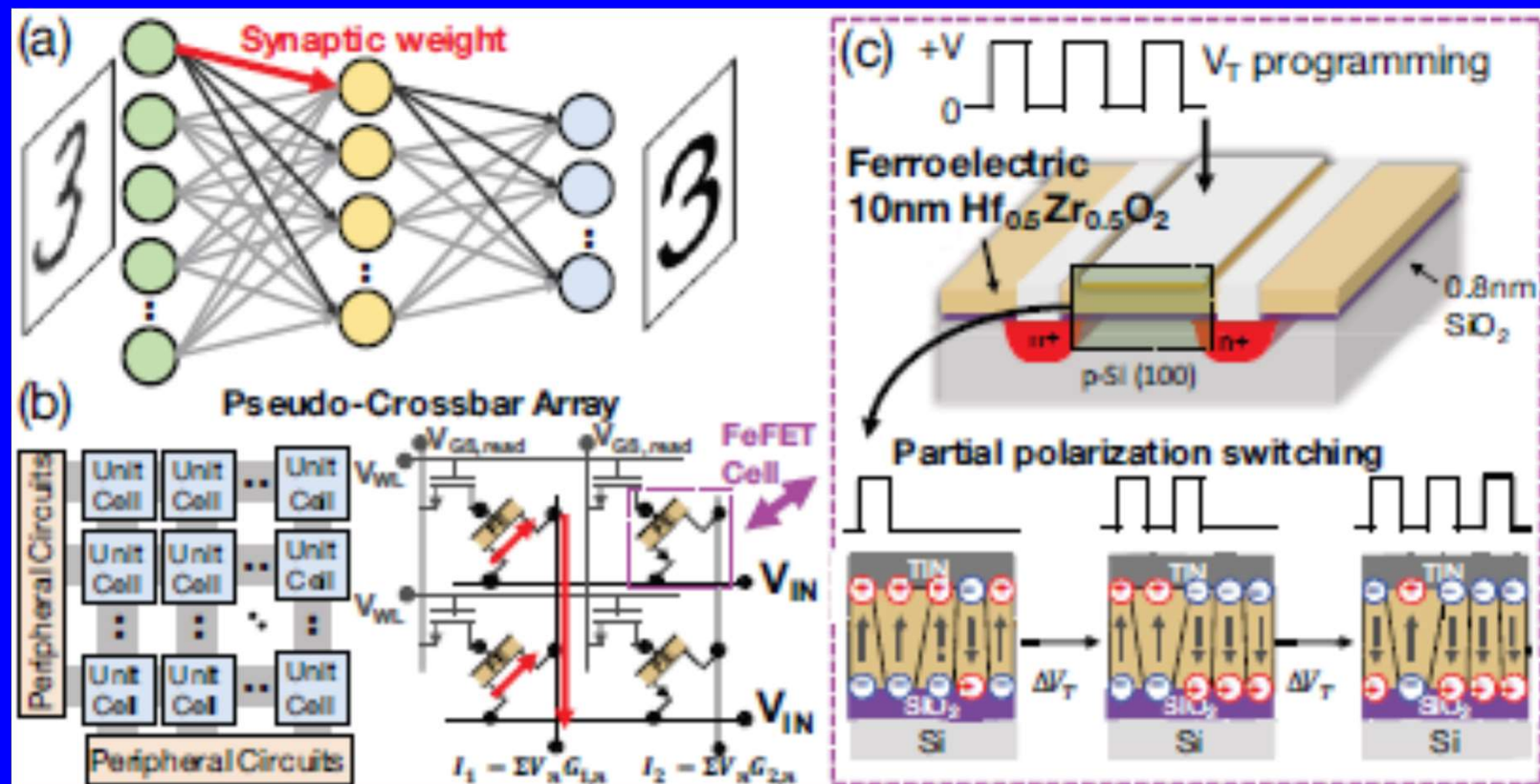


Nanoelectronic technologies and big challenges

**Ferroelectric based FETs /
memories for accelerators
of deep Neural Network
training**

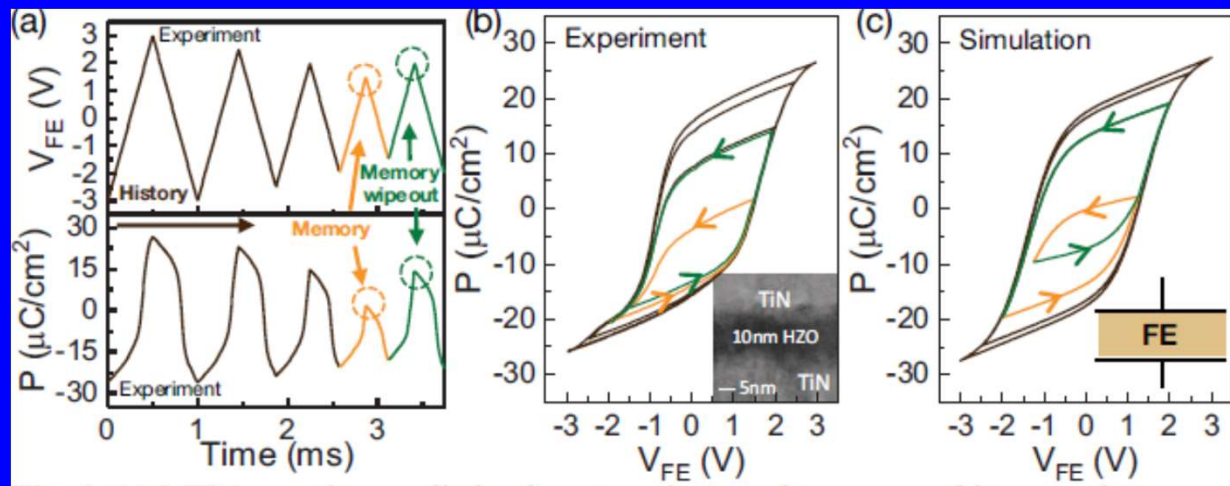
**Rebooting computing:
technology for new
computing paradigms
... quantum computing,
neromorph. computing**

Ferroelectric FET Analog Synapse for Acceleration of Deep Neural Network Training [M. Jerry et al. IEDM 2017]

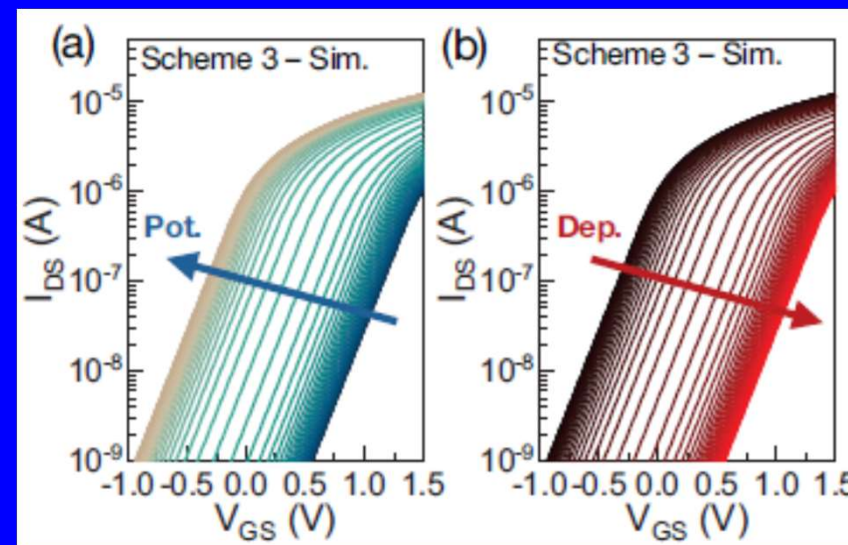


Dispositivi e circuiti elettronici possono realizzare sinapsi e neuroni artificiali e connetterli in *neuromorphic-circuits* e *neuro-morphic-systems* ispirati ai sistemi biologici

Ferroelectric FET Analog Synapse for Acceleration of Deep Neural Network Training [M. Jerry et al. IEDM 2017]



By exploiting minor loops of the ferroelectric one can achieve many discrete values of threshold voltage and device conductance



Nanoelectronic Devices and Circuits with High Energy Efficiency for Internet-of-Things Applications

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13 dicembre 2021

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Progetti per tesi triennali in diversi argomenti

- Transistors with 2D materials for wearable and flexible electronics (collaboration with Aachen (Germany) Stanford (California))
- Ferroelectric based FETs for advanced digital circuits (Negative Capacitance) and neuromorphic, synaptic devices
- Progetto e simulazione di sinapsi artificiali
- Progetto e simulazione di neuroni artificiali

TESI DI LAUREA TRIENNALE

Analisi di Implementazioni Circuitali di Neuroni per Sistemi Neuromorfici

Relatore:

Prof. ESSENI David

Correlatori:

Prof. LOGHI Mirko

Prof. SAGGINI Stefano

Laureando:

NAGELSCHMIED Christian

Anno Accademico 2020/2021

TESI DI LAUREA TRIENNALE

ANALISI DI CIRCUITI PER L'IMPLEMENTAZIONE DELLA PLASTICITA' SINAPTICA

Relatore:

Prof. Mirko Loghi

Laureando:

Simone Saro

Correlatori:

Prof. David Esseni

Prof. Stefano Saggini

ANNO ACCADEMICO 2020/2021

Tesi di Laurea

STUDIO DEL COMPORTAMENTO ANTI-FERROELETTRICO
IN DIELETTRICI IDONEI PER APPLICAZIONI
NANOELETTRONICHE

Relatore:

Prof. David Esseni

Laureando:

Filippo Rupil

Anno accademico: 2020/2021